

Phase 2-Configure DMS Replication from Oracle Hash to Aurora Postgres Hash

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Cleann-up AWS DMS Internal Tables

- **Connect Source and Target**

```
1 #drop "AWSDMS_CONTROL_TABLE" or awsdms_control_schema
2
3 --Audit DB cleanup
4
5 drop table awsdms_control_table.awsdms_apply_exceptions;
6 drop table awsdms_control_table.awsdms_full_load_exceptions;
7 drop table awsdms_control_table.awsdms_history ;
8 drop table awsdms_control_table.awsdms_status ;
9 drop table awsdms_control_table.awsdms_suspended_tables ;
10 drop schema awsdms_control_table;
11
12 ##public
13 drop function public.awsdms_intercept_ddl() cascade;
14 drop table public.awsdms_ddl_audit;
15
16
17
18 --monolith
19 DROP schema "AWSDMS_CONTROL_TABLE";
20 drop schema awsdms_control_schema;
21 #drop awsdms_ddl_audit table
22 drop function awsdms_ddl_audit.awsdms_intercept_ddl() cascade;
23 drop table awsdms_ddl_audit.awsdms_ddl_audit;
24 drop schema awsdms_ddl_audit;
25 ##public
26 drop function public.awsdms_intercept_ddl() cascade;
27 drop table public.awsdms_ddl_audit;
```

Aurora Postgres Hash Database Setup

- **Cluster setup Via IPS**
- Create Aurora Postgres Cluster using IPS Tool , Here Git Config Repo [IPS](#)

```
1 #pre-req deployment
2 cdk list --all -c env=prod-west -c config=psp-prerequisites-prod-west.json
3 cdk diff --all -c env=prod-west -c config=psp-prerequisites-prod-west.json
4 cdk deploy --all -c env=prod-west -c config=psp-prerequisites-prod-west.json
5 #cluster deployment for psp-prod-mon database
6 cdk list --all -c env=prod -c group=primary-mon -c config=psp-prod-mon.json
7 cdk diff --all -c env=prod -c group=primary-mon -c config=psp-prod-mon.json
8 cdk deploy --all -c env=prod -c group=primary-mon -c config=psp-prod-mon.json
```

- **Schema setup**

- Create Schema PSPADM and Required Users, Here Schema and User setup [Scripts](#)

```
1 \i pspadm_tables.sql
2 \i pspadm_hash_part_tables.sql
3 \i pspadm_range_part_tables.sql
4 \i pspadm_const_pk.sql
5 \i pspadm_con_fk.sql
```

DMS Replication Setup (Full-Load + CDC)

• pre-requirement setup on source :

1.Run the following query to verify if supplemental logging is already enabled for the database. Here AWS Docs for [Oracle as source](#)

```
1 SELECT supplemental_log_data_min FROM v$database;
```

2.If not enable , Run the following command to enable supplemental logging at the database level.

```
1 exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
2
3 exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD','PRIMARY KEY');
```

3.Configure CDC using Binary Reader

```
1 Log in to your Amazon RDS for Oracle source database as the master user and run the following stored procedures
  to create the server-level directories.
2 exec rdsadmin.rdsadmin_master_util.create_archivelog_dir;
3 exec rdsadmin.rdsadmin_master_util.create_online_log_dir;
```

4.Create DMS Replication User Source:

```
1 CREATE USER DMS_USER IDENTIFIED by "XXXXXXX"
2     DEFAULT TABLESPACE "PSP_DATA01"
3     TEMPORARY TABLESPACE "TEMP"
4     PROFILE "APPLICATION_HIGH_RISK_PROFILE";
5 GRANT CREATE SESSION to DMS_USER;
6 GRANT SELECT ANY TRANSACTION to DMS_USER;
7 GRANT SELECT on DBA_TABLESPACES to DMS_USER;
8 GRANT EXECUTE on rdsadmin.rdsadmin_util to DMS_USER;
9 GRANT LOGMINING to DMS_USER;
10 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_VIEWS', 'DMS_USER', 'SELECT');
11 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_TAB_PARTITIONS', 'DMS_USER', 'SELECT');
12 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_INDEXES', 'DMS_USER', 'SELECT');
13 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_OBJECTS', 'DMS_USER', 'SELECT');
14 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_TABLES', 'DMS_USER', 'SELECT');
15 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_USERS', 'DMS_USER', 'SELECT');
16 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_CATALOG', 'DMS_USER', 'SELECT');
17 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_CONSTRAINTS', 'DMS_USER', 'SELECT');
18 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_CONS_COLUMNS', 'DMS_USER', 'SELECT');
19 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_TAB_COLS', 'DMS_USER', 'SELECT');
20 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_IND_COLUMNS', 'DMS_USER', 'SELECT');
21 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_LOG_GROUPS', 'DMS_USER', 'SELECT');
22 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$ARCHIVED_LOG', 'DMS_USER', 'SELECT');
23 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOG', 'DMS_USER', 'SELECT');
24 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGFILE', 'DMS_USER', 'SELECT');
25 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$DATABASE', 'DMS_USER', 'SELECT');
26 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$THREAD', 'DMS_USER', 'SELECT');
27 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$PARAMETER', 'DMS_USER', 'SELECT');
28 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$NLS_PARAMETERS', 'DMS_USER', 'SELECT');
29 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$TIMEZONE_NAMES', 'DMS_USER', 'SELECT');
30 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$TRANSACTION', 'DMS_USER', 'SELECT');
31 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$CONTAINERS', 'DMS_USER', 'SELECT');
32 exec rdsadmin.rdsadmin_util.grant_sys_object('DBA_REGISTRY', 'DMS_USER', 'SELECT');
33 exec rdsadmin.rdsadmin_util.grant_sys_object('OBJ$', 'DMS_USER', 'SELECT');
34 exec rdsadmin.rdsadmin_util.grant_sys_object('ALL_ENCRYPTED_COLUMNS', 'DMS_USER', 'SELECT');
35 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGMNR_LOGS', 'DMS_USER', 'SELECT');
36 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$LOGMNR_CONTENTS', 'DMS_USER', 'SELECT');
37 exec rdsadmin.rdsadmin_util.grant_sys_object('DBMS_LOGMNR', 'DMS_USER', 'EXECUTE');
38 -- (as of Oracle versions 12.1 and higher)
39 exec rdsadmin.rdsadmin_util.grant_sys_object('REGISTRY$SQLPATCH', 'DMS_USER', 'SELECT');
40 -- (for Amazon RDS Active Dataguard Standby (ADG))
41 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$STANDBY_LOG', 'DMS_USER', 'SELECT');
42 -- (for transparent data encryption (TDE))
43 exec rdsadmin.rdsadmin_util.grant_sys_object('ENC$', 'DMS_USER', 'SELECT');
44 -- (for validation with LOB columns)
45 exec rdsadmin.rdsadmin_util.grant_sys_object('DBMS_CRYPT0', 'DMS_USER', 'EXECUTE');
46 -- (for binary reader)
47 exec rdsadmin.rdsadmin_util.grant_sys_object('DBA_DIRECTORIES', 'DMS_USER', 'SELECT');
48 -- Required when the source database is Oracle Data guard, and Oracle Standby is used in the latest release of
  DMS version 3.4.6, version 3.4.7, and higher.
49 exec rdsadmin.rdsadmin_util.grant_sys_object('V_$DATAGUARD_STATS', 'DMS_USER', 'SELECT');
```

- **Pre-requirement setup on Target**

- Create DMS Replication User on Target, Here AWS Docs [Postgres as Target](#)

```
1 CREATE USER dms_apg_user WITH PASSWORD 'XXXXXX';
2 GRANT SET ON PARAMETER session_replication_role TO dms_apg_user;
3 GRANT USAGE ON SCHEMA pspadm TO dms_apg_user;
4 GRANT CONNECT ON DATABASE psppp01 TO dms_apg_user;
5 GRANT CREATE ON DATABASE psppp01 TO dms_apg_user;
6 GRANT CREATE ON SCHEMA pspadm TO dms_apg_user;
7 GRANT UPDATE, INSERT, SELECT, DELETE, TRUNCATE ON ALL TABLES IN SCHEMA pspadm TO dms_apg_user;
8 GRANT USAGE ON SCHEMA awsdms_control_schema TO dms_apg_user;
9 GRANT CREATE ON SCHEMA awsdms_control_schema TO dms_apg_user;
10 GRANT UPDATE, INSERT, SELECT, DELETE, TRUNCATE ON ALL TABLES IN SCHEMA awsdms_control_schema TO dms_apg_user;
```

if DMS fail with permission issue for awsdms_ddl_audit . please run below target.

```
1 grant all on public.awsdms_ddl_audit to dms_apg_tgt;
2 grant all on public.awsdms_ddl_audit_c_key_seq to dms_apg_tgt;
```

Endpoint setup source and target (ECA)

- End-Point setup for Oracle Source: AWS DMS, there are two methods for reading the redo logs when doing change data capture (CDC) for Oracle as a source Oracle LogMiner and AWS DMS Binary Reader. LogMiner is an Oracle API to read the online redo logs and archived redo log files. Binary Reader is an AWS DMS method that reads and parses the raw redo log files directly.

```
1 useLogminerReader=N;useBfile=Y
```

- End-Point setup for Postgres Target: For use with change data capture (CDC) only, this attribute has AWS DMS bypass foreign keys and user triggers to reduce the time it takes to bulk load data.Specifies the maximum size (in KB) of any .csv file used to transfer data to PostgreSQL. [Postgres ECA](#)

```
1 afterConnectScript=SET session_replication_role='replica';maxFileSize=1048576;executeTimeout=3600
```

- **DMS Task settings**

- DMS FullLoadSettings: Here AWS Docs For [FullLoad setting](#)

```
1 "FullLoadSettings": {
2     "CommitRate": 50000,
3     "StopTaskCachedChangesApplied": true,
4     "StopTaskCachedChangesNotApplied": false,
5     "MaxFullLoadSubTasks": 16,
6     "TransactionConsistencyTimeout": 600,
7     "CreatePkAfterFullLoad": false,
8     "TargetTablePrepMode": "DO_NOTHING"
```

- DMS LOB Setting: Here AWS Docs For [LOB Setting](#).

Full LOB mode – In full LOB mode AWS DMS migrates all LOBs from source to target regardless of size. In this configuration, AWS DMS has no information about the maximum size of LOBs to expect. Thus, LOBs are migrated one at a time, piece by piece. Full LOB mode can be quite slow.

Limited LOB mode – In limited LOB mode, you set a maximum LOB size for DMS to accept. That enables DMS to pre-allocate memory and load the LOB data in bulk. LOBs that exceed the maximum LOB size are truncated, and a warning is issued to the log file. In limited LOB mode, you can gain significant performance over full LOB mode. We recommend that you use limited LOB mode whenever possible. The maximum recommended value is 102400 KB (100 MB).

```
1 "TargetMetadata": {
2     "ParallelApplyBufferSize": 0,
```

```

3      "ParallelApplyQueuesPerThread": 0,
4      "ParallelApplyThreads": 0,
5      "TargetSchema": "",
6      "InlineLobMaxSize": 0,
7      "ParallelLoadQueuesPerThread": 0,
8      "SupportLobs": true,
9      "LobChunkSize": 64,
10     "TaskRecoveryTableEnabled": false,
11     "ParallelLoadThreads": 0,
12     "LobMaxSize": 0, -----> If limited lob mode true give LOB size KB/MB
13     "BatchApplyEnabled": false,
14     "FullLobMode": true, -----> True to enable Full-lob mode
15     "LimitedSizeLobMode": false, -----> True to limited lob mode
16     "LoadMaxFileSize": 0,
17     "ParallelLoadBufferSize": 0

```

- Error Handling setting: Here [Error Handling_Docs](#)

```

1  "ErrorBehavior": {
2      "FailOnNoTablesCaptured": true,
3      "ApplyErrorUpdatePolicy": "STOP_TASK",
4      "FailOnTransactionConsistencyBreach": false,
5      "RecoverableErrorThrottlingMax": 1800,
6      "DataErrorEscalationPolicy": "STOP_TASK",
7      "ApplyErrorEscalationCount": 10,
8      "RecoverableErrorStopRetryAfterThrottlingMax": false,
9      "RecoverableErrorThrottling": true,
10     "ApplyErrorFailOnTruncationDdl": false,
11     "DataTruncationErrorPolicy": "LOG_ERROR",
12     "ApplyErrorInsertPolicy": "STOP_TASK",
13     "EventErrorPolicy": "IGNORE",
14     "ApplyErrorEscalationPolicy": "STOP_TASK",
15     "RecoverableErrorCount": -1,
16     "DataErrorEscalationCount": 0,
17     "TableErrorEscalationPolicy": "STOP_TASK",
18     "RecoverableErrorInterval": 5,
19     "ApplyErrorDeletePolicy": "STOP_TASK",
20     "TableErrorEscalationCount": 0,
21     "FullLoadIgnoreConflicts": true,
22     "DataErrorPolicy": "STOP_TASK",
23     "TableErrorPolicy": "STOP_TASK"

```

- Change Processing DDL Handling: Here AWS Docs Change [Processing_DDL Handling](#)

```

1  --All parameters are false
2  "ChangeProcessingDdlHandlingPolicy": {
3      "HandleSourceTableDropped": false,
4      "HandleSourceTableTruncated": false,
5      "HandleSourceTableAltered": false

```

• **Table wise DMS Tasks**

- Here Tables info . [PSP - Monolith Database Partitioning](#)

DMS Tasks	TABLE-NAME	Total no.of Tables
Task-1	PSP_LEDGER_BALANCE PSP_MONEY_MOVEMENT_TRANSACTION PSP_ENTRY_DETAIL_RECORD PSP_FINANCIAL_TRANS_STATE PSP_FINANCIAL_TRANSACTION PSP_PAYCHECK PSP_PAYCHECK_SPLIT	7
Task-2	PSP_COMPANY_EVENT PSP_COMPANY_EVENT_DETAIL PSP_COMPANY_EVENT_EMAIL_PARAM	3

Task-3	PSP_COMPENSATION PSP_PROPERTY_AUDIT PSP_QBDT_PAYCHECK_INFO PSP_QBDT_PAYLINE_INFO	4
Task-4	PSP_PSTUB_PAY_ITEM	1
Task-5	PSP_TAX	1
Task-6	PSP_ENTITLEMENT_MESSAGE PSP_PAYCHECK_USAGE PSP_DISBURSE_ADVICE_TAX_LIAB PSP_QBDT_TRANSACTION_INFO PSP_PAYSTUB PSP_DEDUCTION PSP_PSTUB_PAID_TIMEOFF_ITEM PSP_PSTUB_EMPLOYEE_INFO	8
Task-7	Rest of all tables	272
Task-8	PSP_ENTITY_UPDATE	1
Task-9	PSP_TAX_ACCOUNT_AUDIT	1

• **Table Mapping**

- Use table settings to specify any settings that you want to apply to a selected table or view for a specified operation. Table-settings rules are optional, depending on your endpoint and migration requirements. [table mapping](#)

```

1  {
2    "rules": [
3      {
4        "rule-type": "transformation",
5        "rule-id": "1",
6        "rule-name": "1",
7        "rule-target": "column",
8        "object-locator": {
9          "schema-name": "%",
10         "table-name": "%",
11         "column-name": "%"
12       },
13       "rule-action": "convert-lowercase",
14       "value": null,
15       "old-value": null
16     },
17     {
18       "rule-type": "transformation",
19       "rule-id": "2",
20       "rule-name": "2",
21       "rule-target": "table",
22       "object-locator": {
23         "schema-name": "PSPADM",
24         "table-name": "%"
25       },
26       "rule-action": "convert-lowercase",
27       "value": null,
28       "old-value": null
29     },
30     {
31       "rule-type": "transformation",
32       "rule-id": "3",
33       "rule-name": "3",
34       "rule-target": "schema",

```

```

35     "object-locator": {
36         "schema-name": "PSPADM"
37     },
38     "rule-action": "convert-lowercase",
39     "value": null,
40     "old-value": null
41 },
42 {
43     "rule-type": "selection",
44     "rule-id": "16",
45     "rule-name": "16",
46     "object-locator": {
47         "schema-name": "PSPADM",
48         "table-name": "PSP_ARCHIVE_RECORD"
49     },
50     "rule-action": "include",
51     "filters": []
52 }
53 ]
54 }

```

- **Range Boundaries Table Mapping**

- we need to get Range Boundaries from Oracle [DMS Range Boundary sheet](#)
- Below query to get Range Boundaries

```

1  Select /*+ parallel(a,4) */a.COMPENSATION_SEQ From
2      (
3          Select /*+ parallel(e,4) */Row_Number() OVER (Order by COMPENSATION_SEQ) rno, e.COMPENSATION_SEQ
4          From pspadm.PSP_COMPENSATION e order by rno
5      ) a
6  Where rno in
7      (97595388,195190777,292786165,390381553,487976941,585572330,683167718,780763106,878358494,975953883,1073549271,117
8      1144659,1268740047,1366335436,1463930824,1561526212);

```

- **Partition-auto config Table Mapping**

- source more efficiently by loading it in parallel based on all its partitions. [Partition-auto](#)

```

1  {
2      "rule-type": "selection",
3      "rule-id": "4",
4      "rule-name": "4",
5      "object-locator": {
6          "schema-name": "PSPADM",
7          "table-name": "PSP_PSTUB_PAY_ITEM"
8      },
9      "rule-action": "include",
10     "filters": []
11 },
12 {
13     "rule-type": "transformation",
14     "rule-id": "5",
15     "rule-name": "5",
16     "rule-target": "table",
17     "object-locator": {
18         "schema-name": "PSPADM",
19         "table-name": "PSP_PSTUB_PAY_ITEM"
20     },
21     "rule-action": "convert-lowercase",
22     "value": null,
23     "old-value": null
24 },
25 {
26     "rule-type": "table-settings",
27     "rule-id": "12",
28     "rule-name": "12",
29     "object-locator": {
30         "schema-name": "PSPADM",
31         "table-name": "PSP_PSTUB_PAY_ITEM"
32     },
33     "parallel-load": {
34         "type": "partitions-auto"
35     }
36 }

```

Post-Full-Load steps

- **Index creation on Target**

- Once Full-load Completed , then apply Indexes on Target Postgres Database.[Index Scripts](#)

```
1 \i pspadm_Index.sql
2 \i pspadm_hash_create_index.sql
3 \i pspadm_range_create_index.sql
4 \i pspadm_hash_attach_index.sql
5 \i pspadm_range_attach_index.sql
```

Learnings

Problem:Bug, a rare race condition which can cause Postgres relation cache to flush at the same time when a backend process tries to use the relation cache file

```
1 2023-07-12 17:12:27 UTC::@[529]:log: terminating any other active server processes
2 2023-07-12 17:12:27 UTC::@[529]:FATAL: Can't handle storage runtime process crash
3 2023-07-12 17:12:27 UTC::@[529]:log: database system is shut down.
```

Solution: Disable auto vacuum during full load , Instance class is not a variable.

- aurora postgresql versions 14.8 and 15.3 are impacted with the issue. Issue is not specific to any instance class, it could occur on any instance class running affected versions. As updated earlier, unfortunately it is not possible to share an ETA on the fix for the issue.

Problem: invalid input syntax for type integer: "29.0000000000" for table psp_tax_account_audit and column audit_id.

```
1 2023-07-13 13:26:21 UTC:10.95.110.171(52974):dms_apg_user@psppp01:[1413]:error: invalid input syntax for type integer: "29.0000000000"
```

Solution: This error is usually seen when the LOB columns on the target are set as 'Not NULL'. This can also occur when array values are passed to Postgres, but the format of the array may not be correct for target Postgres.

- when you have NUMBER (P,S) on your oracle dms will convert this datatype to REAL8, real8 will support Double datatype in Postgres
- when you set numberDataTypeScale to And precision is greater than 4 and less than or equal to 9, use INT4. and INT4 will convert to int in postgres
- create new task and source endpoint for table and use **numberDataTypeScale=0** in source endpoint add ECA

Problem: value "5000000000007" is out of range for type integer for table psp_quickbooks_info and column token integer

```
1 . value "5000000000007" is out of range for type integer for table psp_quickbooks_info and column token integer
```

Solution: Datatype changed integer to numeric alter table pspadm.psp_quickbooks_info alter column token type NUMERIC(19) ;

Problem: Slow Data Pull/Large Data Set/Heavy Change Rate

```
1 [SOURCE_UNLOAD ]E: ORA-01555: snapshot too old: rollback segment number 15 with name "_SYSSMU15_3465917518$"
too small [1020436] (oracle_endpoint_unload.c:241)
2 01065997: 2023-07-14T05:19:59
```

Solution: Used 24 [logical partitions](#) in place of partition mapping auto for 1 TB+ tables like PSP_COMPANY_EVENT_DETAIL.

- Removed configuration for Full Lob and introduced Limited Lob based on max LOB size for all the long running tasks select /*+parallel(x,8)*/ max(DBMS_LOB.GETLENGTH(changed_attributes)), avg(DBMS_LOB.GETLENGTH(changed_attributes)) FROM pspadm.psp_entity_update x; * Changed DMS Config StreamBufferCount from 3 to 5.

```
1  "TargetMetadata": {
2    "ParallelApplyBufferSize": 0,
3    "ParallelApplyQueuesPerThread": 0,
4    "ParallelApplyThreads": 0,
5    "TargetSchema": "",
6    "InlineLobMaxSize": 0,
7    "ParallelLoadQueuesPerThread": 0,
8    "SupportLobs": true,
9    "LobChunkSize": 64,
10   "TaskRecoveryTableEnabled": false,
11   "ParallelLoadThreads": 0,
12   "LobMaxSize": 16,---->lob max size
13   "BatchApplyEnabled": false,
14   "FullLobMode": false,
15   "LimitedSizeLobMode": true,----> limited lob enabled
16   "LoadMaxFileSize": 0,
17   "ParallelLoadBufferSize": 0
```